

# Supervised Bayesian Matrix Factorization Identifies Prognostic Gene Expression Programs

---

PDAC Project Update — Methods, Validation, and External Benchmarks

Andrew Walther

Department of Biostatistics, UNC Gillings School of Global Public Health

June 18, 2026

# Agenda

**Background** · The PDAC prognostic problem · why supervise the factorization

**Method updates** · The joint model · two linear predictors ( $L\beta$  vs  $(YF)\beta$ ) · explicit priors & non-parametric baseline hazard · the ELBO and its variational family · CAVI & convergence · preprocessing · the cohort-membership extension

**Validation** · Multi-cohort simulation study (completed) · real-data gene programs & survival · external validation on five held-out cohorts

**Impact & Future Directions**

# Background: PDAC needs molecular prognosis

- **Five-year survival < 12%** — among the most lethal solid tumors
- Most patients diagnosed late; **no widely adopted molecular stratification** at diagnosis
- Bulk expression mixes malignant, stromal, and immune signals — prognostic structure is interwoven and often **low-variance**
- **Goal:** recover *gene expression programs* whose activation predicts survival, and validate them on **independent** cohorts

## Why model expression and survival *jointly*?

The common pipeline is **two-step**: run unsupervised factor analysis (PCA / EBMF), then test factors for survival *post hoc*.

- Unsupervised axes track the **largest expression variance** — often batch or housekeeping signal
- A program explaining little variance but strongly prognostic gets **missed**
- **Supervision** steers factors toward survival-relevant structure *during* estimation

*We will demonstrate this with results — not assert it.*

# The model — two coupled components

$$\underbrace{Y = LF^\top + E, \quad E_{ij} \sim N(0, \tau_j^{-1})}_{\text{genomics: low-rank+gene-specific noise}} \quad \underbrace{h_i(t) = h_0(t) \exp(\eta_i)}_{\text{Cox proportional hazards}}$$

| Quantity | Dim          | Meaning                           |
|----------|--------------|-----------------------------------|
| $Y$      | $n \times p$ | expression matrix                 |
| $L$      | $n \times K$ | patient program-activation scores |
| $F$      | $p \times K$ | gene weights per program          |
| $\beta$  | $K \times 1$ | survival (log-HR) coefficients    |
| $\tau_j$ | scalar       | gene-specific precision           |

## Shared loadings couple the two

The **same** latent structure drives low-rank reconstruction of  $Y$  **and** the survival risk score. A program is prognostic only if its  $\beta_k \neq 0$ ; otherwise it explains expression alone.

## Non-parametric baseline hazard

$h_0(t)$  is left **unspecified** — it cancels in the Cox partial likelihood. No Weibull/exponential shape assumption; only proportional hazards.

# Two linear predictors: $L\beta$ vs $(YF)\beta$

## Loadings predictor — $\eta = L\beta$

- Risk uses the **learned** patient scores  $L$  directly
- Survival shapes  $L$  during estimation (fully joint)
- Out-of-sample scoring requires **re-deriving**  $L$  for new patients

## Projection predictor — $\eta = (YF)\beta$

- Risk uses the **direct genomic projection**  $YF$
- **Exact** out-of-sample scoring:  $\eta_{\text{new}} = (Y_{\text{new}}F)\beta$  — the *identical* formula at train and test, no train/test score mismatch

### Priors — every block is explicit

- $q(L_{\cdot k}), q(F_{\cdot k}), q(\beta_k)$ : **point-normal** (spike-and-slab) —  $g(\theta) = \pi_0\delta_0 + (1 - \pi_0)N(0, \sigma^2)$
- Sparsity  $\pi_0$  and slab scale  $\sigma^2$  estimated by empirical Bayes ( **ebnm** ) at each step
- $\tau_j$  (gene precision): **closed-form MLE** update — no EBNM, no prior tuning

### Why we report both

The projection predictor  $(YF)\beta$  gives leakage-free external scoring and is the basis for the recommended model; the loadings predictor  $L\beta$  is the fully-joint reference.



# Joint objective: the ELBO

## Variational family (mean-field, factor-wise)

$$q(L, F, \beta, \tau) = \prod_{k=1}^K q(L_{\cdot k}) \prod_{k=1}^K q(F_{\cdot k}) \prod_{k=1}^K q(\beta_k) \prod_{j=1}^p q(\tau_j)$$

### Genomics term

$$E_q[\mathcal{L}_{\text{gen}}] = \sum_{j=1}^p E_q \log p(Y_{\cdot j} \mid L, F_j, \tau_j)$$

Gaussian reconstruction of  $Y$  by  $LF^\top$ .

### Survival term

$E_q[\mathcal{L}_{\text{Cox}}] \approx$  2nd-order Taylor of the Cox partial log-likelihood

A local quadratic in  $\eta$  makes the Cox term conjugate to the EBNM updates.

## Combined objective

$$\text{ELBO} = \underbrace{E_q[\mathcal{L}_{\text{gen}}]}_{\text{expression fit}} + \underbrace{\lambda E_q[\mathcal{L}_{\text{Cox}}]}_{\text{survival fit}} - \underbrace{\sum_{\text{blocks}} D_{\text{KL}}(q \parallel \pi)}_{\text{prior regularization}}$$

The shared  $L$  enters **both** likelihood terms — that coupling *is* the supervision. ( $\lambda$  rescales the survival term; the recommended fits use  $\lambda = 1$ .) :: {notes} The variational family is mean-field and factor-wise — written out at the top, a direct April request. The ELBO has three pieces: genomics reconstruction, the Cox term via a second-order Taylor expansion that keeps the updates conjugate, and a KL penalty per block. The shared  $L$  in both likelihood terms is the supervision. ::

# Inference: factor-wise CAVI

Coordinate ascent (Gauss–Seidel), per iteration, per factor  $k$ :

$$\beta_k \rightarrow L_{\cdot k} \rightarrow F_{\cdot k} \rightarrow \tau$$

- Each update is an **EBNM** sub-problem: pseudo-data  $x = B/A$ , scale  $s = 1/\sqrt{A}$
- Empirical-Bayes prior  $\hat{g}$  re-estimated **every** update  $\rightarrow$  adaptive shrinkage per factor / per gene
- Deflate the residual after each factor before moving on

**Convergence criterion (both must hold, after burn-in):**

$$\text{mean}_{i,k} |L_{ik}^{(t)} - L_{ik}^{(t-1)}| < 10^{-3}$$

$$\text{and } \text{mean}_k |\beta_k^{(t)} - \beta_k^{(t-1)}| < 10^{-3}$$

## ! Risk-direction sign correction

Cox concordance is invariant to a global sign flip, so an unconstrained fit can land at  $C < 0.5$  (“protective” and “adverse” swapped).

**Fix:** orient each prognostic factor against the **training** concordance, then apply the *same* orientation at test. This removed the earlier sub-0.5 external C-indices.

## i Dual criterion, not max

Averaging over all  $n \times K$  entries (not the max) prevents oscillating factor-orientation boundaries from blocking convergence; requiring **both**  $L$  and  $\beta$  to settle ties genomic structure to survival signal.

# Preprocessing is load-bearing

**The  $p \gg n$  scaling problem.** Merging platforms (RNA-seq, microarray, proteomics) leaves each gene on a different scale. Without correction, between-platform variance swamps the within-platform survival precision  $A_k = \sum_i w_i (Y F)_{ik}^2$  and the survival coefficients **collapse to zero**.

**The fix — normalize *per platform, before merging*:**

1. Log2 transform (RNA-seq only)
2. **Per-cohort gene selection** (combined rank of mean  $\times$  variance), top-3000 each  $\rightarrow$  intersect
3. **Per-platform z-standardization** (each cohort centered/scaled on its own)
4. Row-bind into the merged training matrix

 **Headline: skip per-platform normalization  $\rightarrow \beta$  collapses**

In an 18-configuration benchmark, **10 of 12** pipelines that do **not** z-standardize per platform before merging drive  $\beta \rightarrow 0$  — for *both* linear predictors.

 **No leakage**

Normalization is fit **within each training cohort**, before any split; held-out cohorts get the *same* per-cohort recipe at scoring time.



# Extension: cohort-membership indicator

**Idea.** Append corner-point dummy columns (one per non-reference cohort) to  $L$ , with their survival coefficient **fixed at**  $\beta_{\text{cohort}} \equiv 0$ , so survival risk is driven only by the **biological** factors while the dummies absorb platform offsets in the genomics term.

- Updated via a Normal conjugate step (cohort-row  $F$  update)
- Forcing the dummy in **adds one factor slot** ( $K = 7 \rightarrow 8$ )

## When it helps — and when it doesn't

- At **low K** with weak normalization, the indicator can rescue platform-confounded fits
- Under **per-platform z-standardization**, the offset is *already* removed — the indicator adds a factor slot for **no benefit**

## Result on real data

With per-platform z-std, adding the cohort indicator moves mean external C from **0.636** to **0.614** (and  $K : 7 \rightarrow 8$ ). We therefore recommend **no** cohort indicator.

# Data: synthetic design and real cohorts

## Multi-cohort simulation (completed)

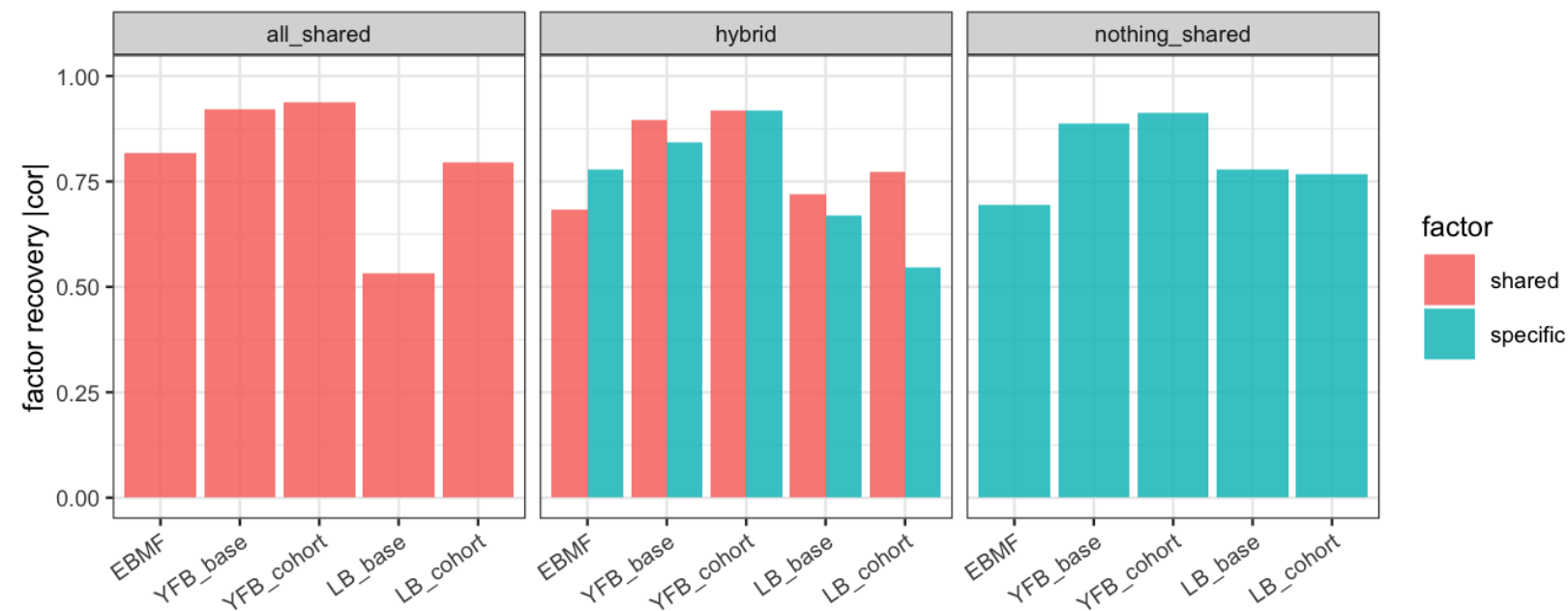
- **EBMF-grounded ground truth:** shared factors (present in all studies) + study-specific factors
- **Shared** factors carry the survival signal; study-specific do not
- **3 scenarios × 5 model arms × 5 seeds**
  - *all-shared*, *hybrid* (realistic), *nothing-shared* (negative control)
- Lets us measure factor **recovery**, **specificity accuracy**, and **C-index** against known truth

## Real PDAC — external validation design

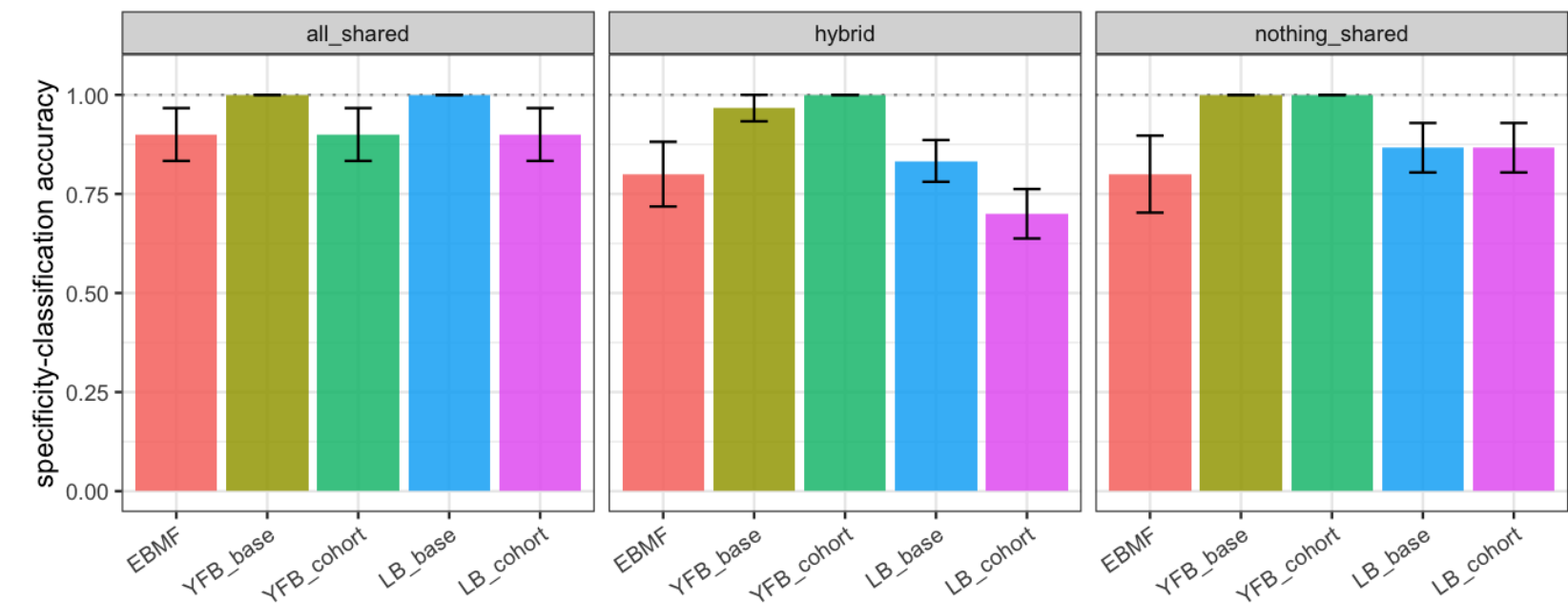
- **Train:** TCGA-PAAD + CPTAC ( $n \approx 273$ , RNA-seq + proteomics)
- **Validate:** 5 fully held-out cohorts, scored by the exact projection formula

| Held-out cohort | Platform   |
|-----------------|------------|
| Dijk            | RNA-seq    |
| Moffitt         | Microarray |
| PACA-AU (array) | Microarray |
| PACA-AU (seq)   | RNA-seq    |
| Puleo           | Microarray |

# Synthetic results I — factor recovery



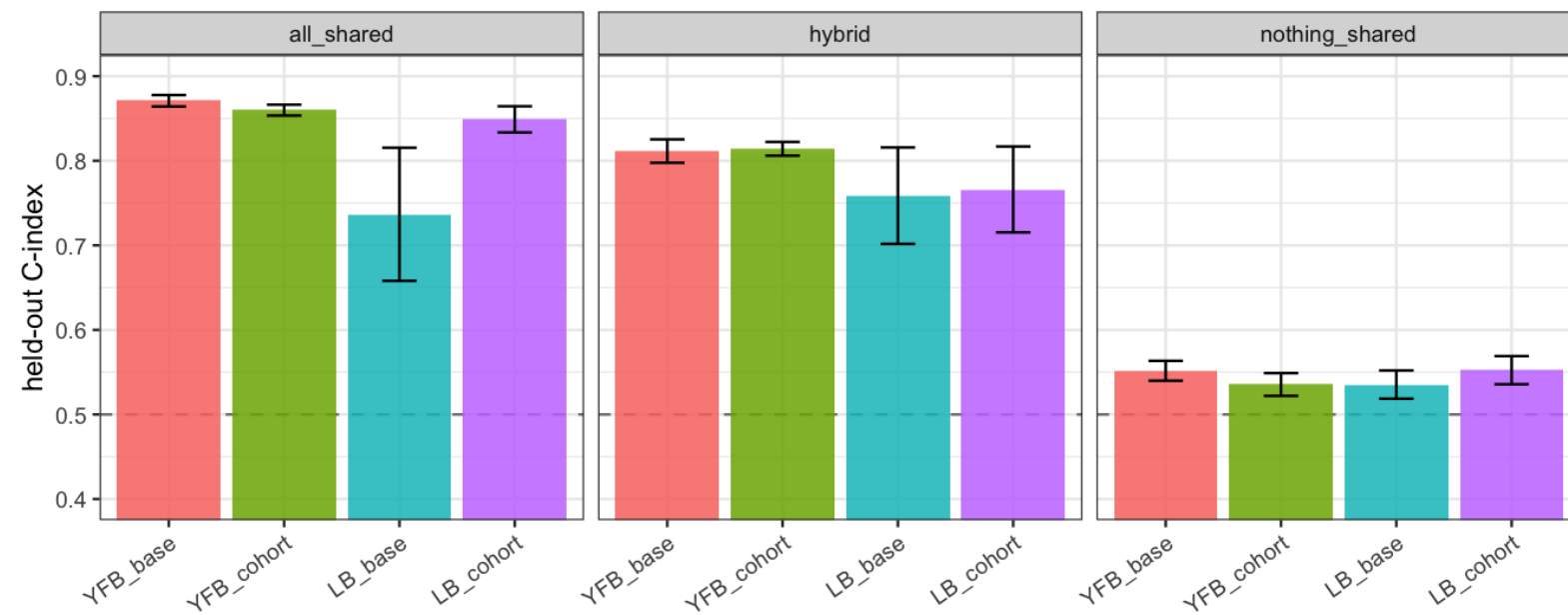
Shared- and study-specific-factor recovery by model arm



Specificity accuracy: shared vs study-specific assignment

- **Shared-factor recovery** for the projection model: **0.92** (all-shared), **0.90** (hybrid) — high fidelity to ground truth
- **Specificity accuracy up to 1.00** — the model natively separates shared from study-specific structure
- This native separation is *why* the cohort-membership indicator adds little: the structure is already recovered

# Synthetic results II — supervised beats two-step



Survival C-index by scenario and model arm (5 seeds)

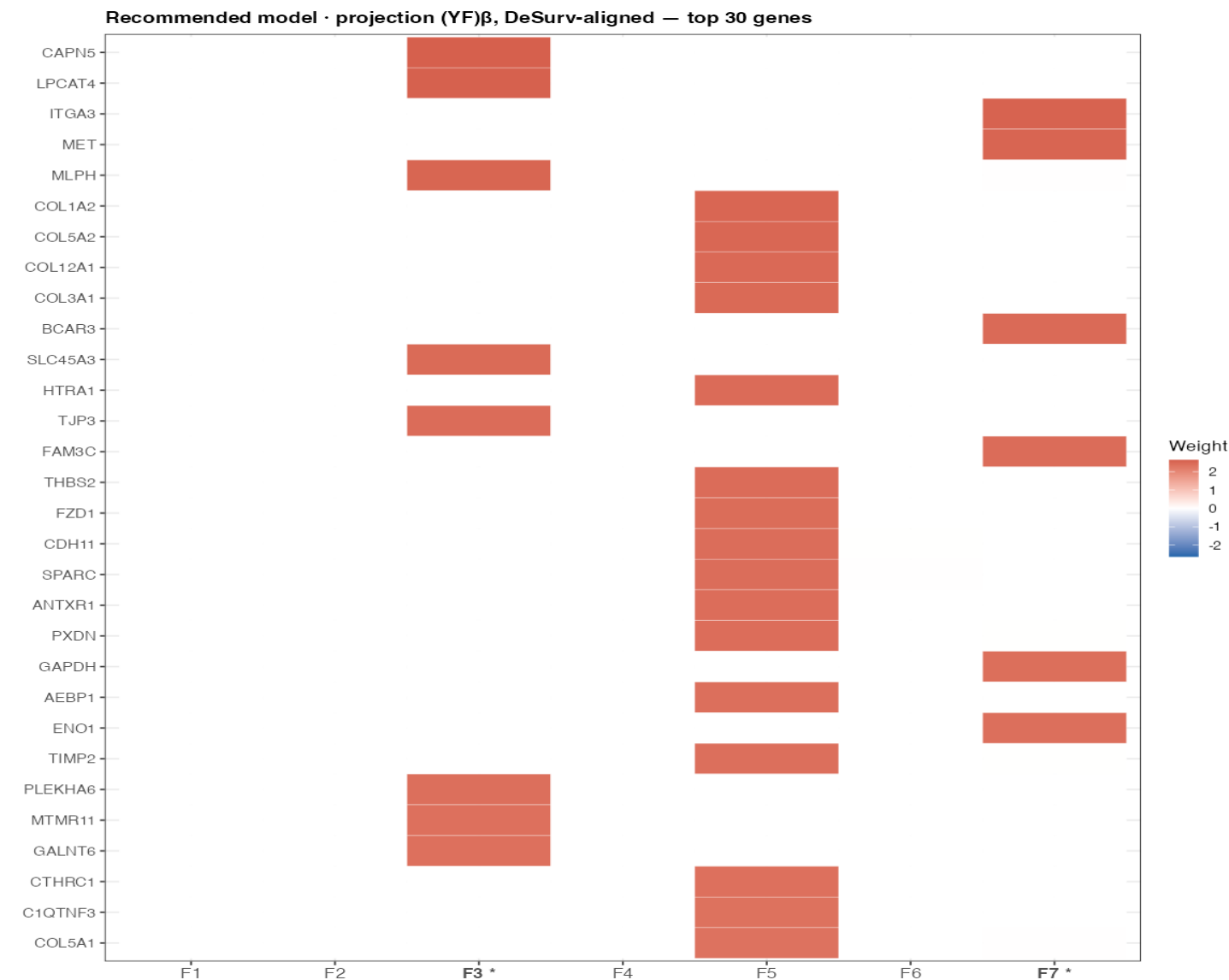
Hybrid (realistic) scenario, mean C-index:

- Projection model (YF) $\beta$ : 0.81
- Loadings model  $L\beta$ : 0.76
- Unsupervised EBMF  $\rightarrow$  Cox: 0.72

**Negative control (nothing-shared):** all arms at  $\approx 0.54$  — no false prognostic signal manufactured when none exists.

*Joint supervision beats the unsupervised two-step on data with known ground truth.*

# Real data I — prognostic gene programs



Gene-program weight heatmap for the recommended model — survival-active programs highlighted

- Recommended model retains  $K = 7$  programs (CV + 1-SE, biological floor  $K \geq 3$ );  $K_{\text{eff}} = 2$  **survival-active** ( $|\hat{\beta}| > 10^{-3}$ )
- One **adverse** program (higher activation → worse survival) and one **protective** program emerge; the remaining programs explain expression only

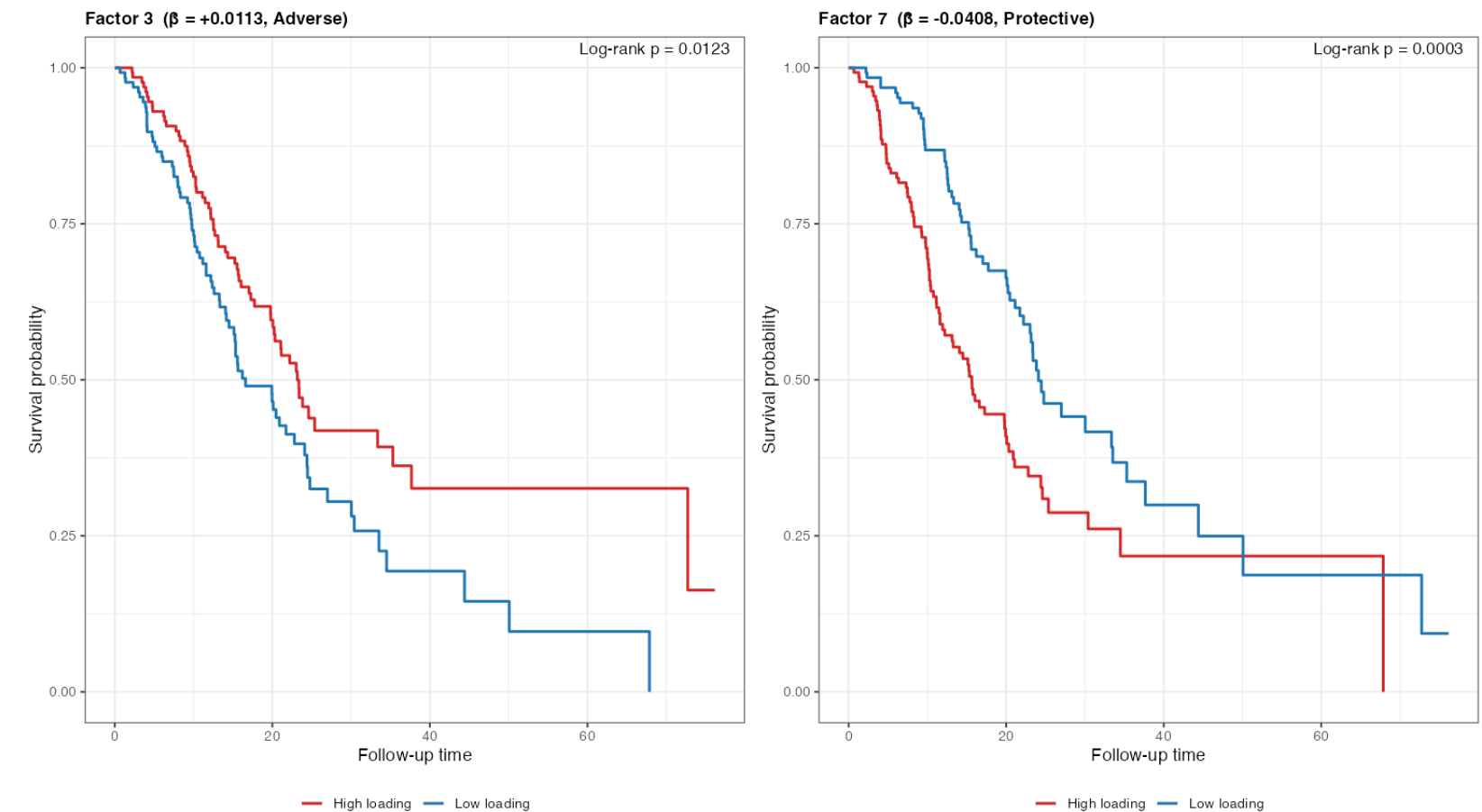


# Real data II — survival stratification

The two survival-active programs stratify patients:

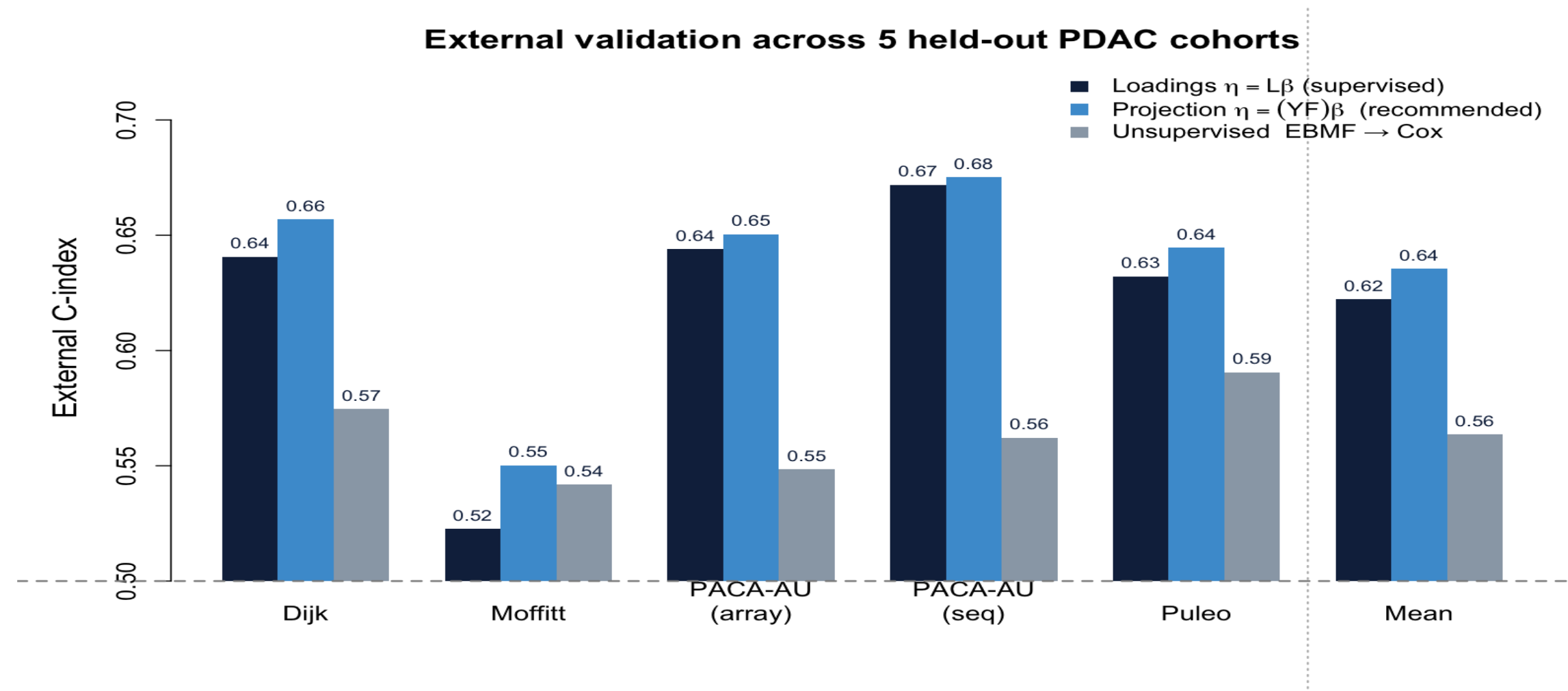
- **Adverse** program: high activation → shorter survival
- **Protective** program: high activation → longer survival
- Log-rank separation in held-in patients; proportional-hazards assumption **not violated** (Schoenfeld test)

Always reported together:  $K = 7$  (CV + 1-SE, floor  $K \geq 3$ ),  $K_{\text{eff}} = 2$ .



*Kaplan-Meier curves by activation of the adverse and protective programs*

# Real data III – external validation, 3-way



External C-index across 5 held-out cohorts: loadings ( $L\beta$ ) vs projection ( $(YF)\beta$ ) vs unsupervised EBMF  $\rightarrow$  Cox

- **Projection model mean C = 0.636** · loadings model 0.622 · EBMF  $\rightarrow$  Cox 0.564, across the five held-out cohorts
- Real-data analog of the synthetic result: the supervised joint model is competitive-to-better than the unsupervised two-step, with  $K = 7$  (CV + 1-SE, floor  $K \geq 3$ )  $\rightarrow K_{\text{eff}} = 2$

# Model comparison summary

| Model                    | Linear predictor       | Preprocessing                         | Cohort ind. | K  | K_eff | Synthetic C | Real ext. C |
|--------------------------|------------------------|---------------------------------------|-------------|----|-------|-------------|-------------|
| Projection (recommended) | $(YF)\beta$            | per-platform z-std + DeSurv selection | no          | 7  | 2     | 0.81        | 0.636       |
| Loadings                 | $L\beta$               | per-platform z-std + DeSurv selection | no          | 7  | 2     | 0.76        | 0.622       |
| Projection + cohort ind. | $(YF)\beta$            | per-platform z-std + DeSurv selection | yes         | 8  | 2     | 0.81        | 0.614       |
| Unsupervised two-step    | EBMF $\rightarrow$ Cox | per-platform z-std + DeSurv selection | n/a         | 20 | n/a   | 0.72        | 0.564       |

Synthetic C = mean held-out C-index, hybrid scenario (5 seeds). Real ext. C = mean across 5 held-out PDAC cohorts. K reported as CV + 1-SE selected with biological floor  $K \geq 3$ ;  $K_{\text{eff}}$  = survival-active factors ( $|\hat{\beta}| > 1e-3$ ).

- **Recommendation:** projection predictor  $(YF)\beta$  · per-platform z-standardization · DeSurv-style gene selection · **no** cohort indicator
- Wins on both axes: best synthetic and real external C-index, and the most **parsimonious, interpretable** survival structure ( $K_{\text{eff}} = 2$ )



# Impact / key findings

## What recent work established

1. **Joint supervision beats the unsupervised two-step** — by ~0.09 C-index on synthetic data with known truth, and competitive-to-better on real held-out cohorts.
2. **Preprocessing is decisive** — per-platform z-standardization before merging is the difference between a working model and  $\beta \rightarrow 0$ ; 10 of 12 alternatives collapse.
3. **Parsimony with signal** — the recommended model carries just  $K_{\text{eff}} = 2$  **survival-active programs** (vs ~20 unsupervised factors) yet matches the field, with **exact, leakage-free external scoring**.

- Mean external C-index = **0.636** across five independent PDAC cohorts
- Competitive with the lab's penalized DeSurv approach, while remaining fully Bayesian (adaptive priors, no  $\alpha$  tuning) and uncertainty-aware

# Future directions & summary

## Next steps

1. **Pathway enrichment** on the two survival-active programs (adverse & protective) — assign biology to the statistical factors
2. **Finalize the EBMF / DeSurv head-to-head** on real data under one shared protocol
3. **Manuscript** — methods-primary framing; PDAC as the application

## Summary

1. Joint supervised factorization with two linear predictors; explicit point-normal priors and non-parametric baseline hazard
2. **Per-platform normalization is load-bearing** — the  $p \gg n$  fix
3. Validated on a completed multi-cohort simulation **and** five external PDAC cohorts
4. Recommended model: projection  $(YF)\beta$ , no cohort indicator,  $K = 7 \rightarrow K_{\text{eff}} = 2$ , mean external C = 0.636

# Discussion

## Thank You

Questions, suggestions, and feedback welcome!

**Andrew Walther** · [awalther@unc.edu](mailto:awalther@unc.edu)

Department of Biostatistics, UNC Gillings School of Global Public Health

# Appendix

---

# Appendix — EBNM pseudo-data & Cox Taylor step

Each CAVI block reduces to an **Empirical-Bayes Normal-Means** problem.

## Normal-means form

- Pseudo-data:  $x = B/A$
- Scale:  $s = 1/\sqrt{A}$
- Model:  $x \mid \theta \sim N(\theta, s^2)$ , prior  $\theta \sim \hat{g}$
- `ebnm` returns posterior mean **and** variance; the **second moment**  $E[\theta^2] = \text{Var} + E[\theta]^2$  is what enters the next update

## Cox term via local quadratic

- The Cox partial log-likelihood is expanded to **second order** in  $\eta$  around the current estimate
- Yields working weights  $w_i$  and a working response, so the survival contribution to  $A$  and  $B$  is **conjugate** to the EBNM update
- Refreshed periodically as  $\eta$  moves

The error-in-variables second moment ( $E[L^2] \neq \bar{L}^2$ ) is what prevents overfitting to uncertain loadings.



# Appendix — $\tau$ update (closed-form MLE)

- $\tau_j$  (gene-specific precision) has **no EBNM step** — it is a closed-form coordinate update:

$$\hat{\tau}_j^{-1} = \frac{1}{n} \sum_{i=1}^n E_q[(Y_{ij} - L_i F_j^\top)^2]$$

- The expectation expands to a variance-corrected residual:

$$\text{Var-term} = E[L^2] E[F^2]^\top - E[L]^2 E[F]^{2\top} \geq 0$$

- Heteroscedastic by design: each gene gets its own precision, down-weighting noisy features rather than treating all genes as equally reliable.

# Appendix — prior & hyperparameter table

| Block         | Prior family | Hyperparameters   | Estimation                           |
|---------------|--------------|-------------------|--------------------------------------|
| $L_{\cdot k}$ | point-normal | $\pi_0, \sigma^2$ | empirical Bayes ( ebnm ), per update |
| $F_k$         | point-normal | $\pi_0, \sigma^2$ | empirical Bayes ( ebnm ), per update |
| $\beta_k$     | point-normal | $\pi_0, \sigma^2$ | empirical Bayes ( ebnm ), per update |
| $\tau_j$      | (none)       | —                 | closed-form MLE coordinate update    |

- $\pi_0$  = spike weight (sparsity);  $\sigma^2$  = slab scale (effect size)
- No user-specified penalty: the prior **learns from the data** at every CAVI iteration
- $\lambda$  (survival-term scale) = 1 in the recommended fits

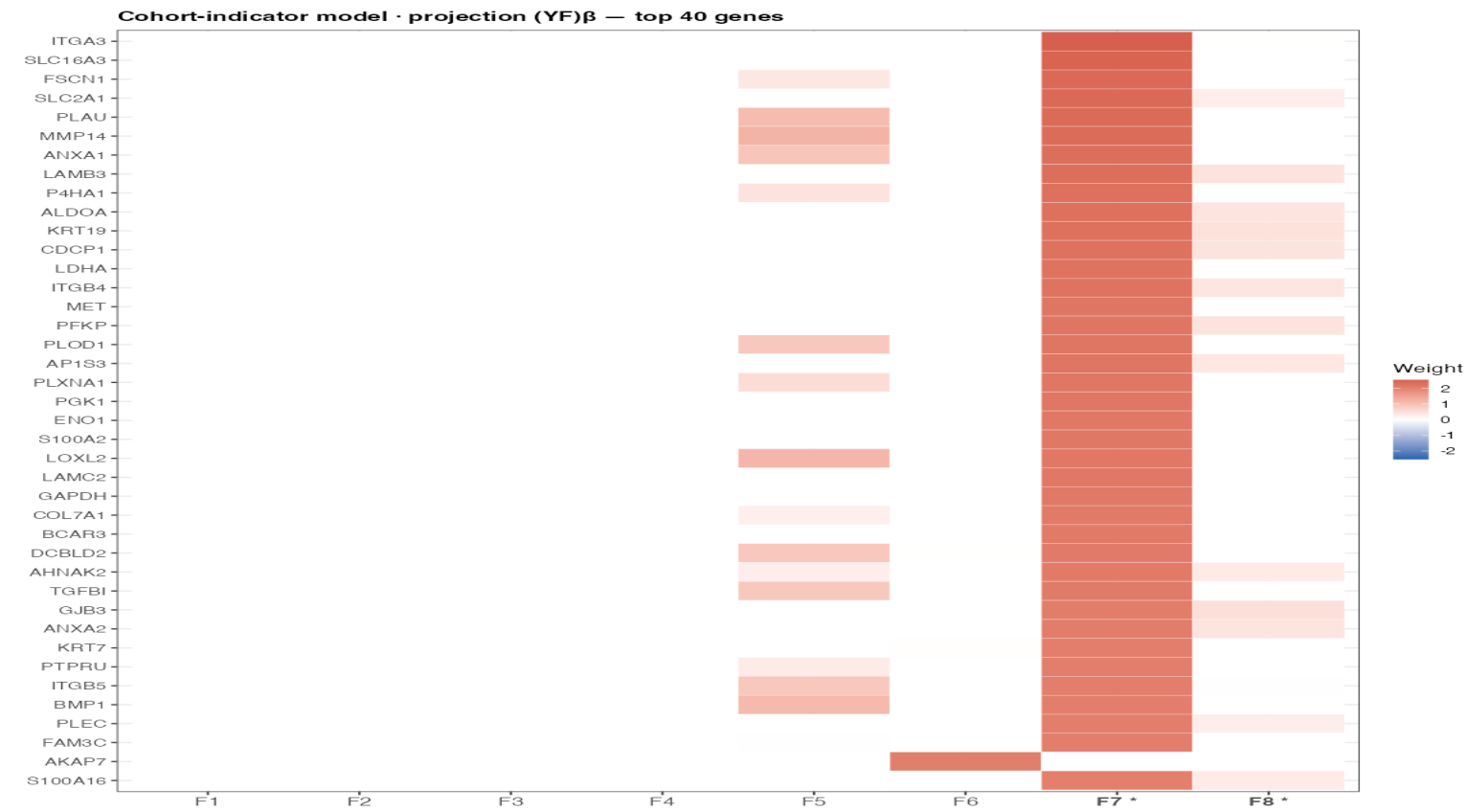
# Appendix — which preprocessing pipelines collapse

| Preprocessing                           | Per-platform z-std before merge? | Outcome                               |
|-----------------------------------------|----------------------------------|---------------------------------------|
| Per-platform z-std + DeSurv selection   | yes                              | $\beta$ recovered (recommended)       |
| Per-platform z-std + variance selection | yes                              | $\beta$ recovered                     |
| Joint quantile + rank                   | no                               | $\beta$ recovered (2 of 12 survivors) |
| Joint z-standardization                 | no                               | $\beta \rightarrow 0$                 |
| Log-transform only                      | no                               | $\beta \rightarrow 0$                 |
| Joint quantile, no rank                 | no                               | $\beta \rightarrow 0$                 |

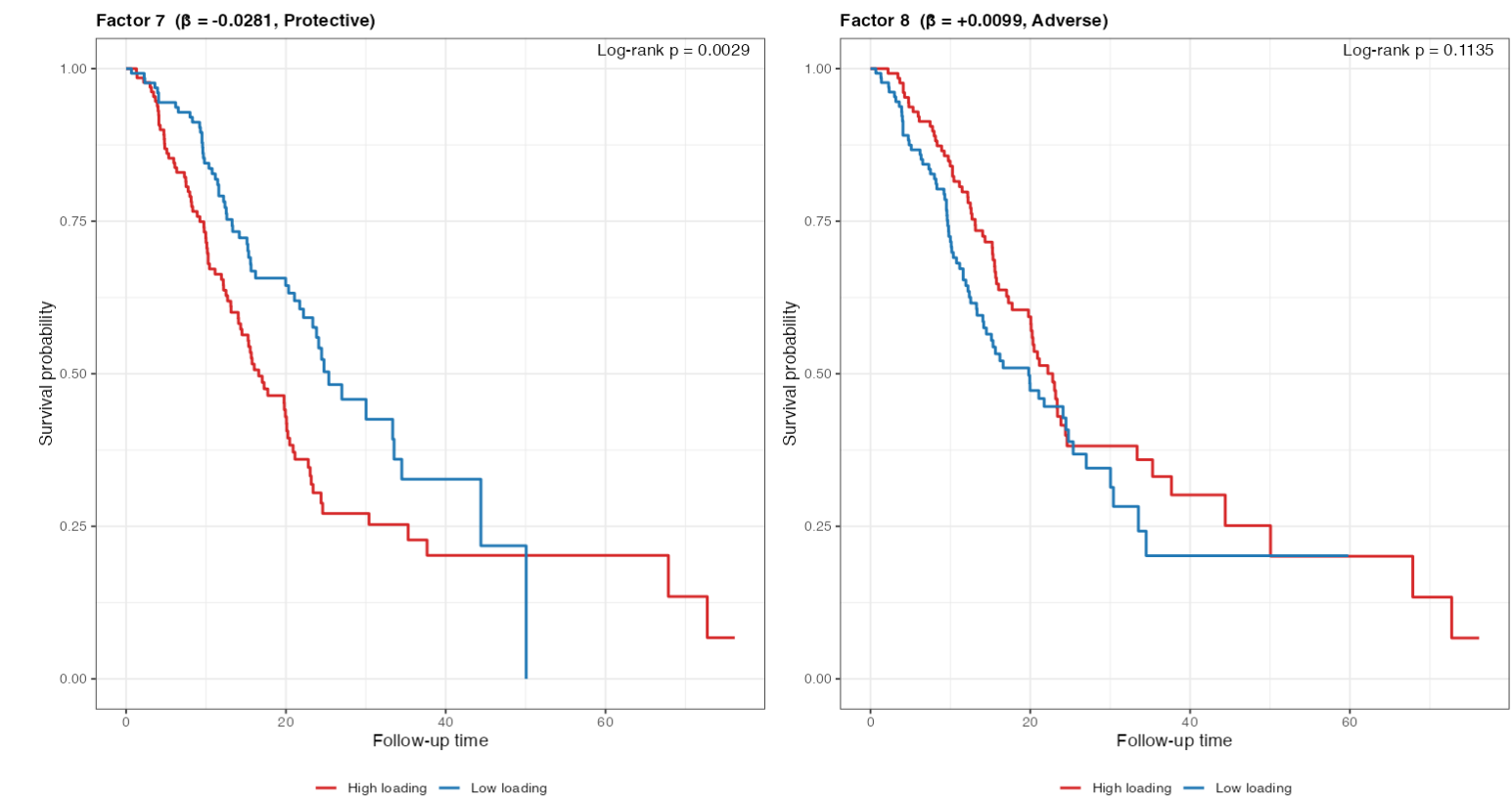
- Across the 18-configuration sweep (5 preprocessing recipes  $\times$  2 linear predictors  $\times$   $\pm$  cohort indicator), **10 of 12** non-per-platform configurations collapse to  $\beta \rightarrow 0$  for both predictors
- The lesson: per-platform normalization is not a tuning detail — it is the enabling step



# Appendix — cohort-indicator contrast



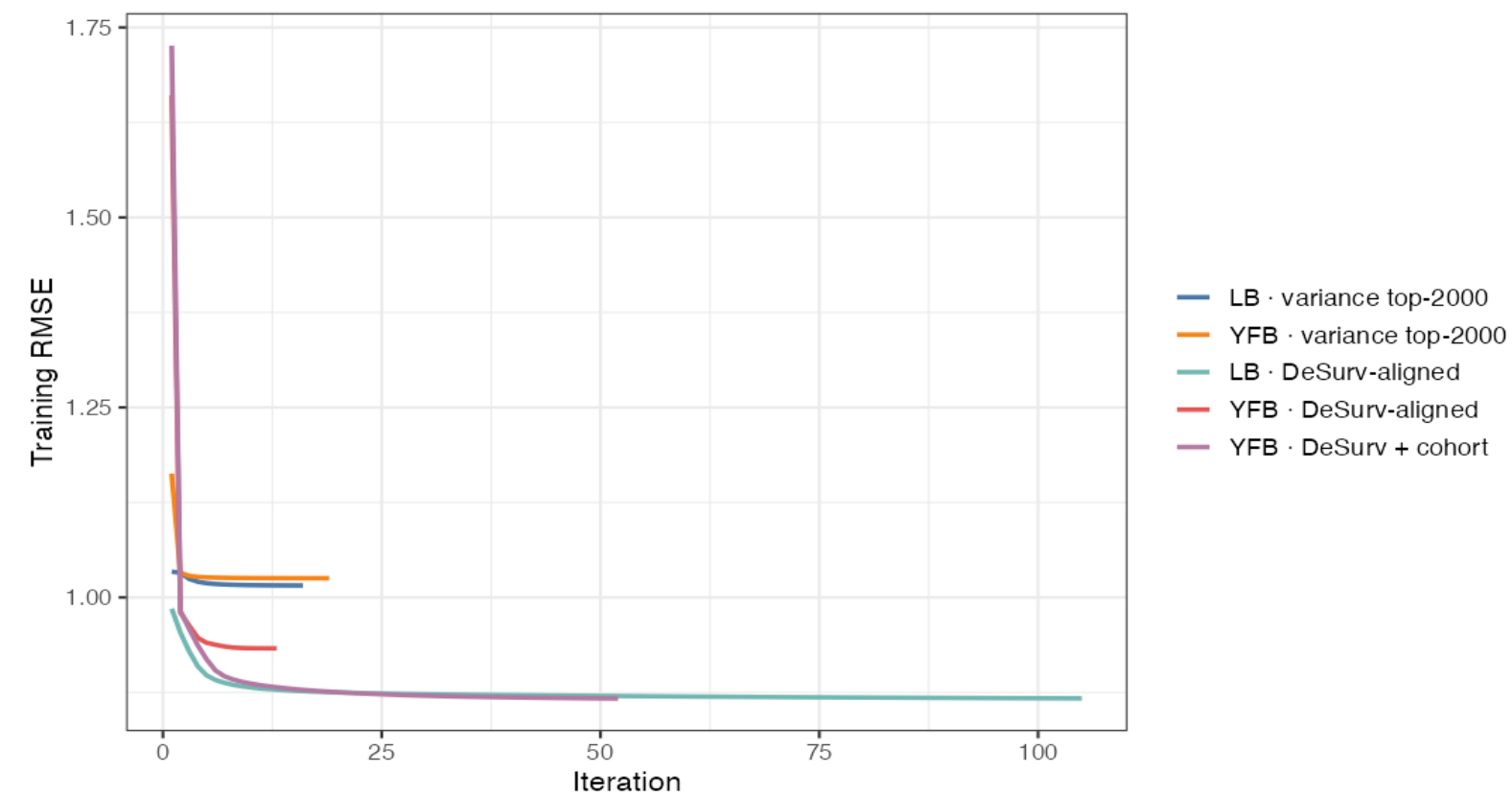
Gene-program heatmap with cohort-membership indicator



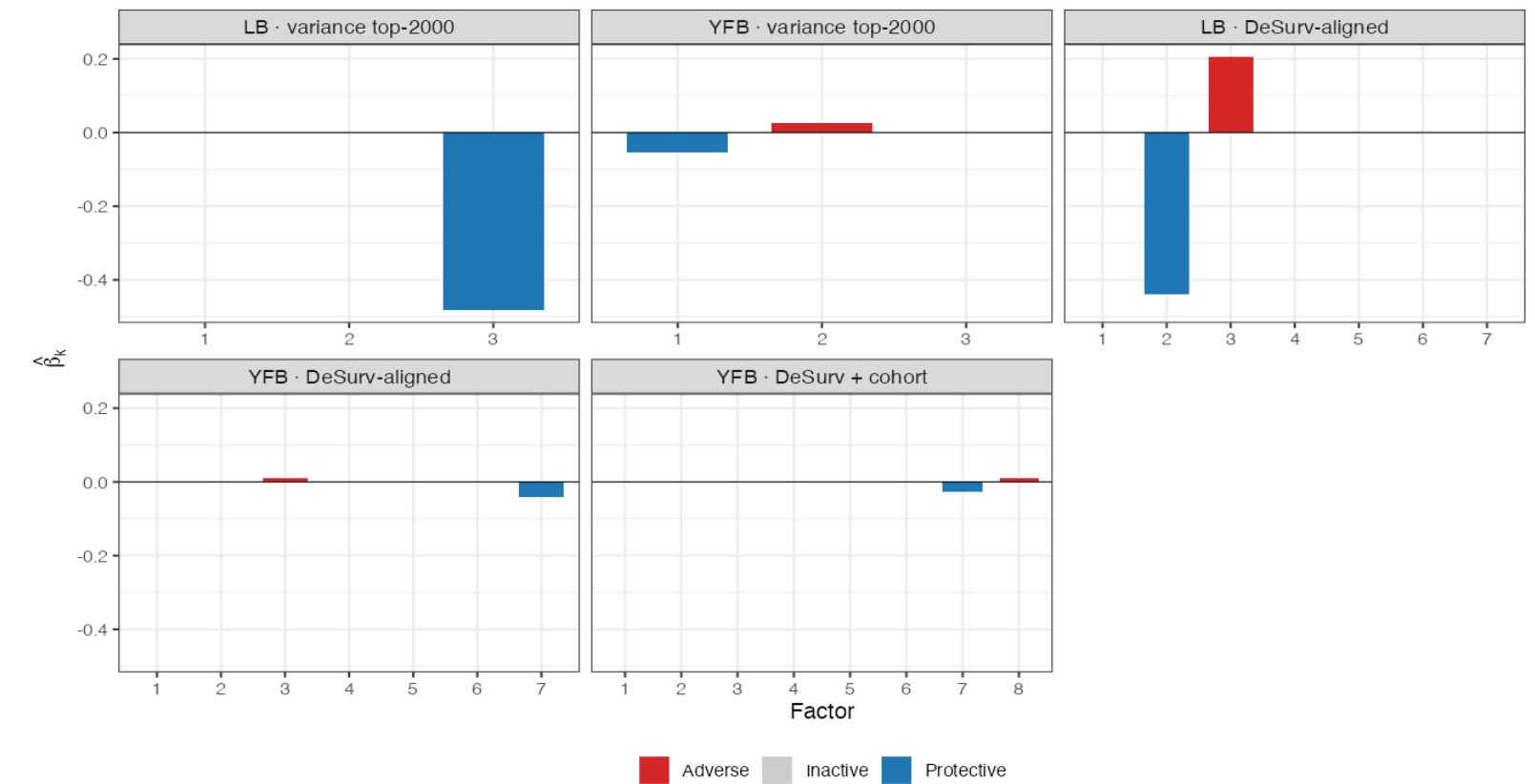
Survival stratification with cohort-membership indicator

- Adding the indicator under per-platform z-std spends a factor slot ( $K : 7 \rightarrow 8$ ) without improving external C-index ( $0.636 \rightarrow 0.614$ )
- The biological survival-active structure is essentially unchanged — confirming the offset is already removed by normalization

# Appendix — convergence & $\beta$ estimates



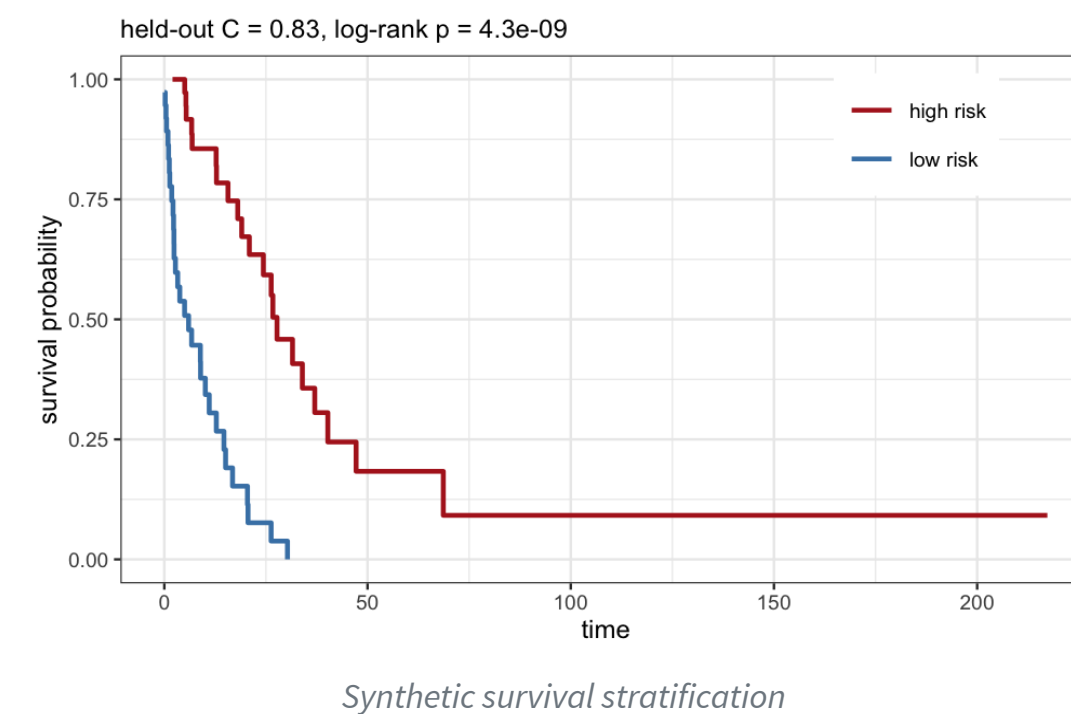
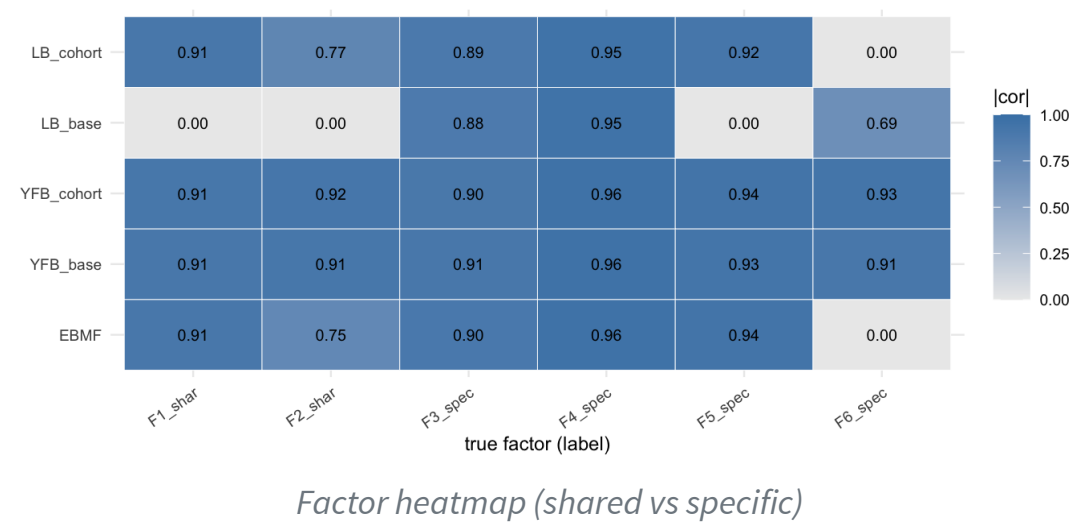
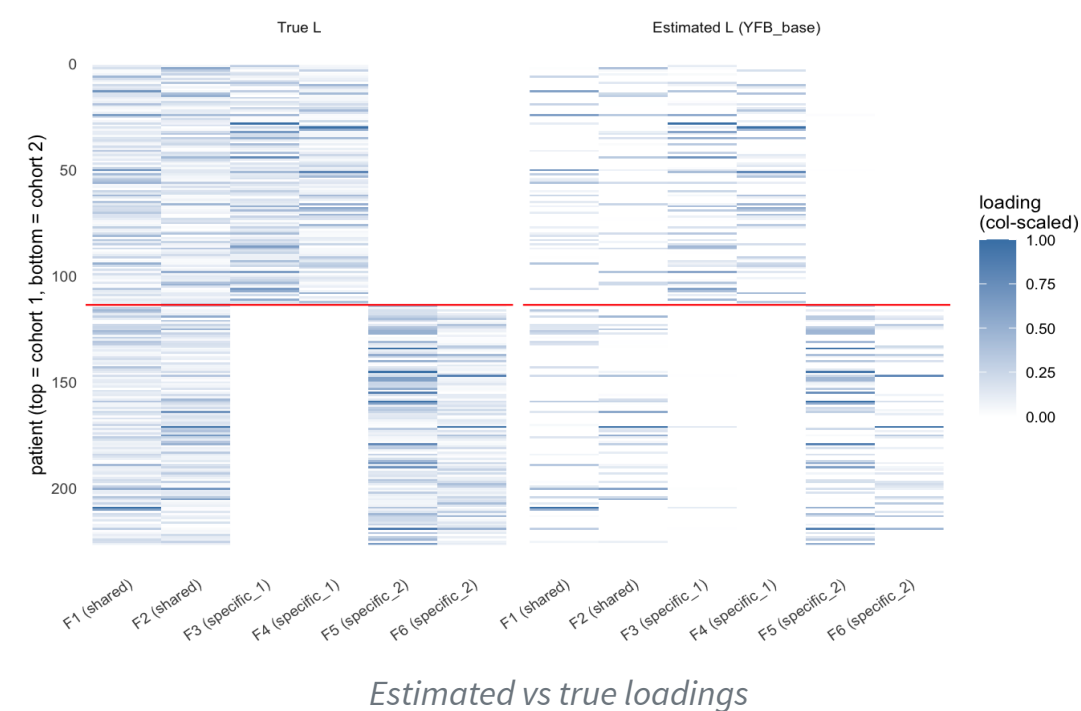
RMSE convergence traces (training fits)



Posterior survival coefficients  $\hat{\beta}$  per program

- RMSE stabilizes to a plateau; the dual  $\Delta L$  &  $\Delta \beta < 1e-3$  criterion governs stopping
- $\hat{\beta}$  barplots show the spike-and-slab at work: most programs shrunk to zero, the survival-active ones retained

# Appendix — synthetic factor structure



- Ground-truth recovery: shared factors (survival-bearing) cleanly separated from study-specific factors
- The simulation establishes that the projection model recovers the *intended* latent structure, not merely a good survival metric